

Syncwright

Documentation

Version 1.0.0

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Source code: https://github.com/RayMunro/Syncwright

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Syncwright

A Mac app that checks whether the sound in a video file lines up with the picture, and corrects it when it does not. It is built for TV episodes and recordings, and it keeps the file's format and encoding as they are wherever that is possible.

- **Checks the whole file**, not just the start. It follows lip movement against speech through the episode, so it works on ordinary dialogue-heavy TV.
- **Fixes without re-encoding.** Video, audio, subtitles and chapters are copied as they are into a new file with the same container. Only the audio timing changes.
- **Offers a re-encode only when needed**, for example when the sync drifts through the episode. Then only the audio is re-encoded, with the same codec, bitrate and channels. The video is never touched.
- **Batch ready.** Drop several episodes or a whole season folder. Each file is checked in turn and "Fix All Needed" repairs the ones that need it.
- **Safe.** Your original is never changed. The corrected copy is saved next to it with `_synced` in the name, and every output is verified and deleted if it is damaged.
- **Plex friendly.** The output keeps the video start time at zero, keeps the MP4 brand and track names, and avoids the extra boxes that stop Plex playing some remuxed MP4 files.

Requirements

- macOS 13 or later.
- ffmpeg and ffprobe. Install them with Homebrew: `brew install ffmpeg`.

Using it

1. Open Syncwright and drop one or more video files, or a folder, on the window. Or choose **Add**.
2. Wait for the check. A file shows "In sync", "Audio X ms late", "Audio X ms early" or "Could not determine".
3. Select a file to see a chart of the audio delay through the episode and how confident the result is.
4. Press **Fix Sync** for that file, or **Fix All Needed** for the whole list.
5. Press **Check Result** on a fixed file to run the check again on the new copy.

See Getting started for the details, How the check works for what the numbers mean, and Fixing sync for what the app writes and when it offers a re-encode.

Limits

- Detection needs dialogue with visible faces, or sound that lines up with movement. Scenes with no faces, heavy music or narration over footage give weak results. The app says "Could not determine" rather than guess.

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- It searches for an offset of up to 3 seconds either way.
- Accuracy is about 40 ms, which is below what most people notice.
- A single shift cannot fix sync that drifts or jumps during a file. For that, the app offers the audio re-encode.

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Getting started

Install

1. Install ffmpeg once: `brew install ffmpeg`.
2. Open the disk image and drag Syncwright to Applications.
3. The first time you open it, right-click the app and choose **Open** if macOS asks you to confirm.

Add files

Drop video files, or a folder, onto the window. Folders are searched for `mkv`, `mp4`, `m4v`, `mov`, `avi`, `ts`, `m2ts`, `mpg`, `mpeg`, `webm` and `wmv` files. Files whose name already ends in `_synced` are skipped. You can also use the **Add** button.

Files are checked one after another. The sidebar shows progress for the file being checked.

Read the result

Select a file to see:

- **The verdict.** "In sync" means the audio is within 40 ms of the picture. Otherwise it says how many milliseconds the audio is late or early.
- **Confidence.** High, Medium or Low. See How the check works.
- **The chart.** Each dot is a section of the file, showing how late the audio is there. Blue dots are sections that matched well. Grey dots did not match and are ignored.
- **A note.** If the sync drifts or jumps during the file, the note says so.

Fix a file

The **Audio delay to remove** box is filled in from the check. You can change it if you know better. Positive means the sound arrives after the picture.

- **Fix Sync** saves a corrected copy without re-encoding.
- **Fix All Needed** does the same for every checked file that is at least 40 ms out and has Medium or High confidence.
- **Fix by Re-encoding Audio** appears when a single shift cannot keep the file in sync. See Fixing sync.

The new file is saved next to the original as `name_synced.ext`. If that name exists, a number is added. Press **Show in Finder** to find it, or **Check Result** to run the check on it.

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Time it takes

The whole file is decoded and every frame is searched for faces, so a feature-length episode takes several minutes. Files are checked one at a time so the Mac stays responsive.

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How the check works

What is measured

Syncwright reads the whole file once and builds two signals at 25 samples a second:

1. **Sound.** The loudness of the dialogue band of the first audio track.
2. **Picture.** How open the mouth is, found with Apple's face landmark detection on every frame. This is compared with speech loudness, because people open their mouths when they talk.

If there are too few faces to be meaningful, or the picture-and-sound method scores better, it falls back to comparing picture movement against sound onsets, which suits claps, cuts and action.

Finding the offset

The two signals are slid against each other across a range of 3 seconds either way. The position where they line up best is the offset. The result is refined to better than one frame.

The file is also split into sections of 4 minutes, and the offset is worked out for each one. That shows whether the sync is steady, drifts steadily, or jumps (for example after an ad break).

The container's own timing is read as well. If the audio track is flagged to start later or earlier than the video, that is added in, so the number you see is the real gap between what you hear and what you see.

Confidence

Confidence is based on how clearly the best offset stands out from the rest, and on how many sections agree.

Level	Meaning
High	A clear match, and most sections agree. Safe to apply.
Medium	A clear match, but with fewer agreeing sections. Usually fine.
Low	A weak match. Check by ear before applying.
None	No reliable relationship found. The app says "Could not determine".

Fix All Needed only touches files with Medium or High confidence.

Patterns

- **Steady.** Every section agrees. A single shift fixes the file.

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- **Drifting.** The offset grows or shrinks through the file, as happens when the audio and video run at slightly different speeds. A single shift cannot fix this.
- **Jumping.** The offset changes between sections because of edits or ad breaks. A single shift fixes only part of the file.

For the last two the app offers the audio re-encode. See Fixing sync.

Accuracy

The measurement is accurate to about 40 ms, one frame at 25 frames a second. People rarely notice errors under about 45 ms when the sound is early and 125 ms when it is late, so the app treats anything under 40 ms as in sync.

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Fixing sync

Without re-encoding

Fix Sync copies every stream into a new file of the same type with new timing. Nothing is decoded or encoded, so there is no loss of quality and it takes seconds.

The video is never moved, so it always starts at time zero. The audio is adjusted instead:

- **Audio is late.** The start of the audio is trimmed by the amount. Audio packets are about 20 to 30 ms long, so the result is within one packet of the target.
- **Audio is early.** The audio is held back by the amount, using the audio track's start time.

All audio tracks move together. Subtitles, chapters and the cover image are kept. Track names and languages are kept.

MP4, M4V and MOV files

These get some extra care, because Plex and some other players are strict about them:

- The file's brand stays `mp42`.
- Timecode and other data tracks are left out. They are not needed for playback.
- The index is moved to the front of the file for fast starts.
- ffmpeg's extra bitrate box inside subtitle tracks is switched off. Newer ffmpeg adds it and Plex's transcoder can reject it.

Verification

After writing, Syncwright checks the new file. It must have the same streams and codecs, a sensible length, and every packet must read cleanly. If any check fails, the new file is deleted and the problem is shown.

With re-encoding, only when needed

If the check finds the sync drifting or jumping, or the lossless fix fails for a file, a **Fix by Re-encoding Audio** button appears.

This re-encodes the **audio only**:

- Same codec, bitrate, sample rate, channel count and language as the original.
- The audio is cut at each section and its speed adjusted very slightly, by well under one percent, so it follows the picture. This is not audible.
- Video, subtitles and chapters are copied untouched, and the container stays the same.

If the audio codec cannot be written by ffmpeg (DTS is one), the audio is written as AAC instead.

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Re-encoded audio is a second generation copy, so there is a small loss of quality. Use it only when the lossless fix cannot do the job.

Output names

The corrected file is saved next to the original as `name_synced.ext`. Your original is never changed or removed.

Plex

Corrected MP4 files play in Plex. If a corrected file still will not play in Plex, check the Plex error and see Troubleshooting.

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Troubleshooting

"ffmpeg was not found"

Install it with Homebrew: `brew install ffmpeg`. The app looks in `/opt/homebrew/bin`, `/usr/local/bin`, `/usr/bin` and `/opt/local/bin`.

"Could not determine sync"

The file had too little to go on. This is common when there are few visible faces, when music or effects cover the dialogue, or for narration over footage. If you know the offset, you can still enter it by hand, but confidence will not be shown.

The result seems wrong

- Check the confidence and the chart. A Low confidence or scattered dots mean you should check by ear.
- Use **Check Result** on a corrected copy. A good fix should show "In sync".
- If the file has several audio tracks, only the first is used for the check. All tracks are corrected by the same amount.

The check is slow

Every frame is searched for faces, so a long file takes several minutes. Close other heavy apps, and note that files are checked one at a time.

A corrected file will not play in Plex

- Confirm the original plays in Plex. If it does not, the problem is not the fix.
- Compare the two files with `ffprobe` for differences such as extra streams.
- Try the corrected file in another player. If it plays there but not in Plex, send the Plex error text when you report the problem.

The lossless fix failed verification

The app deletes a damaged output and says why. Try **Fix by Re-encoding Audio**, which rebuilds the audio stream.

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macOS will not open the app

Right-click Syncwright and choose **Open**, then confirm. This is needed once for apps that are not notarised.

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Building from source

Syncwright is a Swift Package that builds a SwiftUI macOS app.

Requirements

- macOS 13 or later, with Xcode or the Xcode command line tools (Swift 5.9 or later).
- ffmpeg and ffprobe at run time: `brew install ffmpeg`.
- Google Chrome, to render the documentation PDF.

Building the app

```
./build_app.sh
open Syncwright.app
```

`build_app.sh` builds a release binary, assembles Syncwright.app with its icon and Info.plist, and signs it with your Apple Development certificate if you have one (otherwise ad hoc).

Building the disk image

```
./make_dmg.sh
```

This builds the app, renders the documentation PDF, and creates `dist/Syncwright-<version>.dmg` containing the app, an Applications shortcut, the documentation, the guides and the licence.

Testing from the command line

The app binary has a headless mode for testing:

```
Syncwright.app/Contents/MacOS/Syncwright --analyze video.mp4
Syncwright.app/Contents/MacOS/Syncwright --analyze video.mp4 --fix out.mp4 120
Syncwright.app/Contents/MacOS/Syncwright --analyze video.mp4 --reencode out.mp4
```

The first prints the measured offset. The second writes a corrected copy with the audio delay given in milliseconds. The third writes an audio re-encoded copy. Set `SYNCWRIGHT_DEBUG=1` for per-section details.

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Other tools

Tool	Purpose
`tools/make_icon.swift`	Draws the app icon into `Resources/AppIcon-1024.png`
`tools/make_docs_pdf.py`	Renders the documentation chapters into `docs/Syncwright-Documentation.pdf`
`tools/make_dmg_bg.swift`	Draws the disk image window background
`tools/make_dmg_icon.swift`	Draws the disk image's own icon
`tools/set_file_icon.swift`	Gives the finished `.dmg` file that icon in Finder

Project layout

File	Purpose
`Sources/Syncwright/Analyzer.swift`	Reads the file and measures the offset
`Sources/Syncwright/Fixer.swift`	Writes the corrected file and verifies it
`Sources/Syncwright/Tools.swift`	Runs ffmpeg and ffprobe
`Sources/Syncwright/ContentView.swift`	The window
`Sources/Syncwright/App.swift`	App entry point and test mode

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Changelog

1.0.0 (November 2026)

- First release.
- Checks sync across a whole video file using lip movement against speech, with a picture motion fallback.
- Lossless fix that never moves the video and keeps the container, codecs, subtitles and chapters.
- Audio only re-encode for files whose sync drifts or jumps.
- Batch checking and fixing of episodes and folders.
- Verifies every corrected file, and removes it if it is damaged.
- MP4 output tuned for Plex.

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